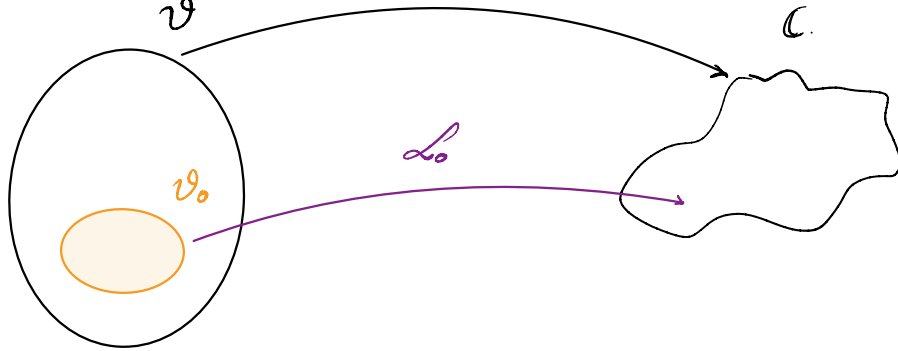


Clase 6 Análisis Numérico

Teorema de aproximación de Weierstrass: Toda función continua en un compacto puede ser aproximada uniformemente por polinomios. i.e., si $f \in C(I)$ y $\varepsilon > 0$ entonces $\exists p \in \mathcal{P}$ t.q. $\|f - p\|_{\infty} < \varepsilon$.

Teorema de Hahn-Banach: Sea V un espacio normado complejo y $L_0: V_0 \rightarrow \mathbb{C}$ es continua sobre el subespacio $V_0 \subseteq V$ entonces $\exists L: V \rightarrow \mathbb{C}$ t.q. $L|_{V_0} = L_0$ y $\|L\| = \|L_0\|$.



Recordemos que

$$\begin{aligned}\|L\| &= \inf \{c \mid |Lx| \leq c\|x\|\} \\ &= \{c \mid |L(\frac{x}{\|x\|})| \leq c\} \\ &= \sup_{\|x\|=1} \{|Lx|\}.\end{aligned}$$

(Spanning Criterion): Sea W un subespacio de V donde V es normado entonces $v \in \bar{W}$ si y solo si $\forall L \in W^\perp$ $L(v) = 0$.

$$W^\perp = \{L: W \rightarrow \mathbb{C} \text{ continuous} \mid L(w) = 0 \ \forall w \in W\}.$$

Proof. $[\Rightarrow]$. Sea $v \in \bar{W}$ y $L \in W^\perp$ es decir $L|_W \equiv 0$. Sea $\varepsilon > 0$. entonces existe $w \in W$ t.q. $\|v - w\| < \varepsilon$.

Entonces

$$\begin{aligned}\|L(v)\| &= \|L(v - w)\| \\ &\leq \|L\| \|v - w\| \\ &< \|L\| \varepsilon.\end{aligned}$$

$$\begin{aligned}\|L\| &= \inf \{c \mid \|Lx\| \leq c\|x\|\} \\ &= \sup \left\{ \frac{\|Lx\|}{\|x\|} ; x \neq 0 \right\} \\ &\Rightarrow \|Lx\| \leq \|L\| \|x\|.\end{aligned}$$

dado que L es continua i.e., $\|L\| < \infty$ y $\varepsilon > 0$ arbitrario, $L(v) = 0$.

[$\Leftarrow$]. Por contrareciproco:

$$v \notin \bar{W} \text{ entonces } \exists L \in W^\perp \text{ } L(v) \neq 0$$

Supongamos que $v \notin \bar{W}$ entonces $d(v, W) > 0$. Definamos:

$$V_0 = \{av + w \mid w \in W, a \in \mathbb{C}\}. \text{ es subespacio de } V.$$

y definamos

$$L_0: V_0 \longrightarrow \mathbb{C}.$$
$$av + w \longmapsto a$$

$$\text{Note que } L_0|_W \equiv 0.$$

Claim: L_0 es continuo.

$$\|L\| = \inf \{C \mid |Lx| \leq \|x\|C\}.$$

L es acotado si $\exists K \|Lx\| \leq K\|x\|$

$$\|av + w\| \geq |a| d(v, W) \text{ entonces } |a| \leq \frac{\|av + w\|}{d(v, W)} \text{ entonces.}$$

$$|L_0(av + w)| = |a| \leq \|av + w\| \cdot C \quad \text{donde } C = \frac{1}{d(v, W)}$$

por definici3n L_0 es acotado y por tanto continuo.

Por thm Hahn-Banach $\exists L: V \rightarrow \mathbb{C}$ t.q. $L|_{V_0} = L_0$ y $\|L\| = \|L_0\|$.

$$L|_W = L_0|_W \equiv 0 \text{ es decir } L \in W^\perp \text{ y } L(v) = L_0(v) = 1 \neq 0. \quad \square$$

Proof: (Thm de Weierstrass): (WAT). Es suficiente probar el teorema

para $f(x) = |x|$ for $-1 \leq x \leq 1$. Sea. $V = (C[-1, 1], \|\cdot\|)$. y.

$$W = \{\text{polinomios}\}$$

Goal: $|x| \in \bar{W}$ Por spanning criterion $\forall L \in W^\perp \quad L(f) = 0$.

$$f(x) = |x|.$$

Sea $L \in W^\perp$ vamos a probar que $L(|x|) = 0$.

$$\text{Sea } |t|^2, \quad z = x + iy \text{ y } x > 0 \text{ entonces } |t|^2 = e^{z \ln |t|}$$

y consideremos $f(z) := L(|t|^2)$ queremos que $f(1) = 0$.

Lemma: $f(x, y) = L(|t|^{x+iy})$ es holomorfa en $U = \{z \mid \operatorname{Re} z > 0\}$.

$$(f \text{ es holomorfa si } \frac{\partial f}{\partial \bar{z}} = 0).$$

Proof: Lemma: Definimos $g(x, y, t) = |t|^{x+iy} = e^{(x+iy) \ln |t|}$

$$g: \mathcal{U} \times \underbrace{[-1, 1]}_{K} \longrightarrow \mathbb{C} \quad \text{entonces} \quad g_{x,y}: [-1, 1] \longrightarrow \mathbb{C}.$$

$$f(x, y) = \mathcal{L}(g_{x,y}).$$

Derivando:

$$\left. \begin{aligned} \frac{\partial}{\partial x} (g) &= |t|^{x+iy} \ln |t| \\ \frac{\partial}{\partial y} (g) &= i |t|^{x+iy} \ln |t| \end{aligned} \right\} \text{son continuas:}$$

entonces:

$$\frac{\partial f}{\partial x} = \mathcal{L}\left(\frac{\partial}{\partial x} |t|^{x+iy}\right) = \mathcal{L}(|t|^{x+iy} \ln |t|)$$

$$\frac{\partial f}{\partial y} = \mathcal{L}\left(\frac{\partial}{\partial y} |t|^{x+iy}\right) = i \mathcal{L}(|t|^{x+iy} \ln |t|)$$

Así

$$\frac{\partial f}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right) = 0. \quad \text{Así } f \text{ es holomorfa en } \mathcal{U} \quad \blacksquare$$

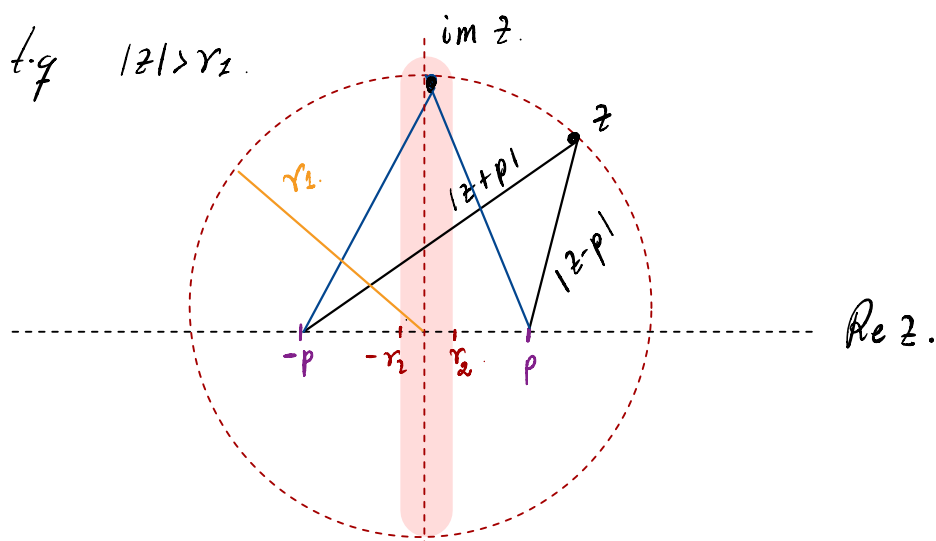
Dado que $f(z_n) = 0$ entonces:

$$\frac{f(z)}{z-2}, \frac{f(z)}{z-4}, \dots, \frac{f(z)}{z-2n} \quad \text{son todas holomorfas:}$$

Lemma: Sea $p \in \mathbb{R}$, $p > 0$ y $\varepsilon > 0$ entonces $\exists r_1, r_2 > 0$ t.q.

$$\left| \frac{z+p}{z-p} \right| \leq 1 + \varepsilon. \quad \forall z \in \mathbb{C} \quad \text{t.q.} \quad |z| > r_1.$$

$$\circ \quad |\operatorname{Re} z| < r_2$$



Consideremos

$$\frac{z+1}{z-1} \cdot \frac{z+2}{z-2} \cdots \frac{z+N}{z-N} f(z) \text{ es holomorfa en } U.$$

Entonces $\forall \varepsilon > 0 \exists r_1, r_2 > 0$ s.t. $1 \leq k \leq N \quad \left| \frac{z+k}{z-k} \right| \leq 1 + \varepsilon$ para $|z| > r_1$
o $|\operatorname{Re} z| < r_2$.

$$\left| \frac{z+1}{z-1} \cdots \frac{z+N}{z-N} \right| |f(z)| \leq (1+\varepsilon)^N |f(z)|$$

Así:

$$|f(z)| = |L(1/z)| \leq \|L\| \|1/z\|$$

$$= \|L\| \max\{|t|^{-2}; -1 \leq t \leq 1\} = \|L\|.$$

así:

$$\left| \frac{z+1}{z-1} \cdots \frac{z+N}{z-N} \right| |f(z)| \leq \underbrace{(1+\varepsilon)^N}_{\varepsilon \rightarrow 0} \|L\| \leq \|L\|.$$

Note que

$$\left| \frac{z+k}{z-k} \right| = \left| 1 + \frac{2z}{k-z} \right| \text{ entonces}$$

$$\left| 1 + \frac{2z}{z-k} \right| \cdots \left| 1 + \frac{2z}{z-N} \right| |f(z)| \leq \|L\| (1+\varepsilon)^N \leq \|L\|.$$

Si $z=1$.

$$|f(1)| \left(1 + \frac{2}{1}\right) \left(1 + \frac{2}{3}\right) \cdots \left(1 + \frac{2}{2N-1}\right) \leq \|L\| (1+\varepsilon)^N \leq \|L\|$$

Entonces:

$$\left(1 + \sum_{i=1}^N \frac{2}{2i-1}\right) |f(1)| \leq \left(1 + \frac{2}{1}\right) \left(1 + \frac{2}{3}\right) \cdots \left(1 + \frac{2}{2N-1}\right) |f(1)|$$

$$\leq \|L\|$$

tomando $N \rightarrow \infty$. $\sum_{i=1}^{\infty} \frac{2}{2i-1} = \infty$. entonces $f(1) = 0$.

lo que prueba que $|L(1/z)| = 0$.

Maximum - Modulus thm.: Sea $U \subseteq \mathbb{C}$ conjunto abierto y acotado y f holomorfa en un conjunto abierto que contiene a U . Si $|f(z)| \leq C$.
 $\forall z \in \partial U$ entonces $|f(z)| \leq C \quad \forall z \in U$.

i.e., $\forall z \in U \quad |f(z)| \leq \max \{ |f(z)| \mid z \in \partial U \}.$

Problema: $\forall \{a_i\}$ sucesión de no negativos y $\forall n \in \mathbb{N}$.

$$1 + \sum_{i=1}^n a_i \leq \prod_{i=1}^n (1 + a_i).$$

Thm: Sea $\{\Phi_0, \Phi_1, \dots, \Phi_n, \dots\}$ es un conjunto de polinomios ortogonales. Entonces $\exists (\alpha_k)_{k=1}^\infty, (\beta_k)_{k=1}^\infty, (\gamma_k)_{k=1}^\infty$ tal que se cumple la fórmula de recurrencia

$$\Phi_{k+1}(x) = \alpha_k(x - \beta_k)\Phi_k(x) - \gamma_k\Phi_{k-1}(x).$$

Proof: Note que $x\Phi_k \in \Pi_{k+1}$ entonces $\{\Phi_0, \Phi_1, \dots, \Phi_{k+1}\}$ es base ortogonal. así $\left\{ \frac{\Phi_0}{\|\Phi_0\|}, \frac{\Phi_1}{\|\Phi_1\|}, \dots, \frac{\Phi_{k+1}}{\|\Phi_{k+1}\|} \right\}$ es base ortonormal.

$$\begin{aligned} x\Phi_k &= \sum_{i=0}^{k+1} \left\langle x\Phi_k, \frac{\Phi_i}{\|\Phi_i\|} \right\rangle \cdot \frac{\Phi_i(x)}{\|\Phi_i\|} \\ &= \sum_{i=0}^{k+1} \frac{\langle x\Phi_k, \Phi_i \rangle}{\|\Phi_i\|^2} \Phi_i(x) \end{aligned}$$

Note que $x\Phi_i \in \Pi_{i+1} \Rightarrow \sigma_{k,i} = 0$ para todo $i=0, 1, \dots, k-2$.

$$x\Phi_k(x) = \frac{\langle x\Phi_k, \Phi_{k+1} \rangle}{\|\Phi_{k+1}\|^2} \Phi_{k+1}(x) + \frac{\langle x\Phi_k, \Phi_k \rangle}{\|\Phi_k\|^2} \Phi_k(x) + \frac{\langle x\Phi_k, \Phi_{k-1} \rangle}{\|\Phi_{k-1}\|^2} \Phi_{k-1}(x)$$

Lemma: Suponga que $\{\Phi_0, \Phi_1, \dots, \Phi_k, \dots\}$ son polinomios ortogonales para $\langle \cdot, \cdot \rangle$. Entonces $\langle \Phi_k, q \rangle = 0$ $q \in \Pi_{k-1}$. $\{ \text{degree}(\Phi_i) \leq i \ \forall i \}$.

$$\Phi_0 = 1, \quad \Phi_{k+1} = x^{k+1} - \sum_{i=0}^k \frac{\langle x^{k+1}, \Phi_i \rangle}{\|\Phi_i\|^2} \Phi_i$$

Veamos que $\langle x\Phi_k, \Phi_j \rangle = 0$ si $j=0, 1, \dots, k-2$.

$$\begin{aligned}
\langle x \Phi_k, \Phi_j \rangle &= \left\langle x \Phi_k, x^j - \sum_{i=0}^{j-1} \frac{\langle x^j, \Phi_i \rangle}{\|\Phi_i\|^2} \Phi_i \right\rangle \\
&= \langle x \Phi_k, x^j \rangle - \sum_{i=0}^{j-1} \frac{1}{\|\Phi_i\|^2} \langle x \Phi_k, \underbrace{\langle x^j, \Phi_i \rangle \Phi_i}_{\#} \rangle \\
&= \langle x \Phi_k, x^j \rangle - \sum_{i=0}^{j-1} \frac{\langle x^j, \Phi_i \rangle}{\|\Phi_i\|^2} \langle x \Phi_k, \Phi_i \rangle.
\end{aligned}$$

fijando x .

$$\begin{aligned}
\langle x \Phi_k, \Phi_{k-2} \rangle &= \left\langle x \left(x^k - \sum_{j=0}^{k-1} \frac{\langle x^k, \Phi_j \rangle}{\|\Phi_j\|^2} \Phi_j \right), \Phi_{k-2} \right\rangle \\
&= \left\langle x^{k+1} - \sum_{j=0}^{k-1} \frac{\langle x^{k+1}, \Phi_j \rangle}{\|\Phi_j\|^2} \Phi_j, \Phi_{k-2} \right\rangle \\
&= \underbrace{\left\langle x^{k+1} - \sum_{j=0}^k \frac{\langle x^{k+1}, \Phi_j \rangle}{\|\Phi_j\|^2} \Phi_j, \Phi_{k-2} \right\rangle}_{\Phi_{k+1}} + \frac{\langle x^{k+1}, \Phi_k \rangle}{\|\Phi_k\|^2} \langle \Phi_k, \Phi_{k-2} \rangle.
\end{aligned}$$

$$= \langle \Phi_{k+1}, \Phi_{k-2} \rangle + \frac{\langle x^{k+1}, \Phi_k \rangle}{\|\Phi_k\|^2} \langle \Phi_k, \Phi_{k-2} \rangle = 0.$$

Entonces:

$$x \Phi_k(x) = \frac{\langle x \Phi_k, \Phi_{k+1} \rangle}{\|\Phi_{k+1}\|^2} \Phi_{k+1}(x) + \frac{\langle x \Phi_k, \Phi_k \rangle}{\|\Phi_k\|^2} \Phi_k(x) + \frac{\langle x \Phi_k, \Phi_{k-1} \rangle}{\|\Phi_{k-1}\|^2} \Phi_{k-1}(x)$$

Implica que

$$\begin{aligned}
\Phi_{k+1}(x) &= \frac{\|\Phi_{k+1}\|^2}{\langle x \Phi_k, \Phi_{k+1} \rangle} \left\{ x - \frac{\langle x \Phi_k, \Phi_k \rangle}{\|\Phi_k\|^2} \right\} \Phi_k(x) \\
&\quad - \frac{\langle x \Phi_k, \Phi_{k-1} \rangle}{\langle x \Phi_k, \Phi_{k+1} \rangle} \frac{\|\Phi_{k+1}\|^2}{\|\Phi_{k-1}\|^2} \Phi_{k-1}(x)
\end{aligned}$$

$$\text{Sea } \alpha_k = \frac{\|\Phi_{k+1}\|^2}{\langle x \Phi_k, \Phi_{k+1} \rangle} \quad \beta_k = \frac{\langle x \Phi_k, \Phi_k \rangle}{\|\Phi_k\|^2}, \quad \gamma_k = \frac{\langle x \Phi_k, \Phi_{k-1} \rangle}{\langle x \Phi_k, \Phi_{k+1} \rangle} \frac{\|\Phi_{k+1}\|^2}{\|\Phi_{k-1}\|^2}.$$

